MATH 115A: LINEAR ALGEBRA
2024 SPRING - LECTURE 6
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MIDTERM EXAMINATION 1

Your Name

The Last **Six**

By signing below, you confirm that you did not cheat on this exam.

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INSTRUCTIONS

• Your submission is due 4:00 PM on Gradescope.

• Please allow sufficient time for uploading your submission (at least 7 minutes).

• If you have a question at any time during the exam, please join our usual Zoom link, and send me a private message in the chat.

• Question 5 is extra credit. You can earn the full 200 points without attempting Question 5. Any points you earn from Question 5 will be added to the points you earn from Questions 1-4, up to a total of 200. That is, even with extra credit, your score cannot exceed the maximum of 200.

• Make sure to show all your work and justify your answers fully.

• Make sure to write legibly. **Illegible work will not be graded.**

• If you are using a pencil, make sure your writing is dark and easily visible.

• Any work you cross out will not be graded.

FRIENDS, GOOD LUCK! YOU'VE GOT THIS!

1. (50 pts) Let $F = (F, +, \cdot)$ be a field. Show that for all $a \in F$,

$$0a = 0.$$

In your proof, use the following principles **only** to justify each step; you may refer to them by the numbers below. Some of the principles may get used more than once, while others may not get used at all.

(1) Definition and existence of 0 in the group $(F, +)$

(2) Definition and existence of 1 in the group $(F^\times, \cdot)$

(3) Definition and existence of $-a$ for all $a \in (F, +)$

(4) Definition and existence of a^{-1} for all $a \in (F^\times, \cdot)$

(5) Associativity in $(F, +)$

(6) Associativity in $(F^\times, \cdot)$

(7) Commutativity of the group $(F, +)$

(8) Commutativity of the group $(F^\times, \cdot)$

(9) Distributivity in $(F, +, \cdot)$

Here, let me get you started:

Proof. For all $a \in F$, we have

$$0a = 0a \qquad | \text{ by using (1)}$$

$$(0+0)a = 0a \qquad |$$

$0 \cdot a = 0 \cdot a$ | by (1)

$(0+0)a = 0a$ | by (9)

$0a + 0a = 0a$ | by (1)

$0a + 0a = 0a + 0$ | by (3)

$0a + 0a + (-0a) = 0a + 0 + (-0a)$ | by (7)

$0a + 0a + (-0a) = 0 + 0a + (-0a)$ | by (5)

$0a + (0a + (-0a)) = 0 + (0a + (-0a))$ | by (3)

$0a + 0 = 0 + 0$ | by (1)

$\therefore \ 0a = 0$

2. (*a*) (10 pts) Complete the following statement:

A nonempty subset W of a vector space V is a subspace if and only if the following two conditions hold:

(1) closure under $+$: $\forall\ x, y \in W$, $x+y \in W$

(2) closure under $\cdot$: $\forall\ a \in F, x \in W$, $a \cdot x \in W$

(*b*) (40 pts) For the rest of this question, we will consider the vector space

$$M_{2\times 2}(\mathbb{F}_3) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mid a, b, c, d \in \mathbb{F}_3 \right\},$$

where $\mathbb{F}_3$ is the finite field $\{0, 1, 2\}$, with addition and multiplication defined modulo 3 (see **Lesson 4**, pp.10-11).

For each of the following four sets, decide whether the set is a subspace of $M_{2\times 2}(\mathbb{F}_3)$ and prove your claim by using your answer to part (*a*).

(1) $\left\{ \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix} \right\} = A$

Closure under $+$: $\begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix} \oplus \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 3 \\ 3 & 0 \end{pmatrix} \bmod 3 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \notin A$

$\forall\ x, y$ in subspace, $x+y \in A$

$\therefore$ Not a subspace

(2) $\left\{ \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix} \right\} = B$

closure under $+$: $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \oplus \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix} \in B$

$\begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix} \oplus \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \in B$

closure under $\cdot$: $\alpha = \{0, 1, 2\}$ $\qquad 0 \cdot \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix} \in B$

$\forall\ \alpha \cdot \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \in B$, $\qquad 1 \cdot \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix} \in B$

$\forall\ \alpha \cdot \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix} \in B$ $\qquad 2 \cdot \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix} \in B$

$\therefore$ closure under $+$ & $\cdot$ so it is a subspace

(Question 2(b) continued)

(3) $\left\{\begin{pmatrix} 0 & b \\ 0 & 0 \end{pmatrix} \mid b \in \mathbb{F}_3\right\} \cup \left\{\begin{pmatrix} 0 & 0 \\ c & 0 \end{pmatrix} \mid c \in \mathbb{F}_3\right\} = C$

Closure under $+$: $\begin{pmatrix} 0 & a \\ 0 & 0 \end{pmatrix} \oplus \begin{pmatrix} 0 & 0 \\ b & 0 \end{pmatrix} = \begin{pmatrix} 0 & a \\ b & 0 \end{pmatrix} \notin C$

$\therefore$ not a subspace

(4) $\left\{\begin{pmatrix} 0 & b \\ 0 & 0 \end{pmatrix} \mid b \in \mathbb{F}_3\right\} \cap \left\{\begin{pmatrix} 0 & 0 \\ c & 0 \end{pmatrix} \mid c \in \mathbb{F}_3\right\} = D$

$D = \left\{\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}\right\}$

$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \oplus \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \in D$

$0 \cdot \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \in D$

$1 \cdot \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \in D$

$2 \cdot \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \in D$

$\therefore$ It is a subspace as its closed under both.

3. (*a*) (10 pts) Let V be a vector space over $\mathbb{R}$. State what it means for the set $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\} \subseteq V$ to be *linearly independent.*

let $\{u,v,w\} \subseteq V$. The set is linearly independent if $\exists$ vectors

$x_1, \ldots, x_n \in \{u, v, w\}$ & $a_1, \ldots, a_n \in \mathbb{R}$ s.t.

$a_1x_1 + \cdots + a_nx_n = 0$ only if

$a_j = 0$ where $1 \leq j \leq n$

(*b*) (40 pts) Let V be a vector space over $\mathbb{R}$. Suppose that $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\} \subseteq V$ is a linearly independent set. Show that the set $\{\mathbf{u}+\mathbf{v}, \mathbf{u}+\mathbf{w}, \mathbf{v}+\mathbf{w}\}$ is linearly independent.

$a(u+v) + b(u+w) + c(v+w) = \vec{0}$

$u(a+b) + v(a+c) + w(b+c) = \vec{0}$

$a+b=0 \Rightarrow a+(-c)=0 \Rightarrow a=c$

$a+c=0 \Rightarrow a+a=0 \Rightarrow 2a=0 \Rightarrow a=0$

$b+c=0 \Rightarrow b+c-c=0-c$ $\quad b=0$

$b+0=-c$ $\quad c=0$

$b=-c$

$\therefore$ the $\{u+v, u+w, v+w\}$ is lin ind

4. (50 pts) Let V be a finite-dimensional vector space over a field F, and let S be a finite linearly independent set in V. Prove that S can be extended to a basis of V. If in your argument you are using any prior theorems, make sure to indicate this clearly (you do not need to give full statements of the prior theorems you use; a clear indication that a prior result is being used is enough).

$\because$ V is finite dimensions, $\dim_F(V) < \infty$

since S is lineasly independent $|S| \leq \dim_F(V)$, according to the replacement thm

· If $|S| = \dim_F(V)$, then it'll be a lin ind set, aka a basis for V

· If $|S| < \dim_F(V)$, adjoin vectors from $V \setminus S$ s.t. new set K is linearly independent

We know that no lin ind set in V can have more elements than $\dim_F(V)$ so this repitition of adding vectors to K must end after a finite number of steps, giving us a lin ind set to which no further elements from V can be added.

let $\bar{x} \in V$, $\bar{x} \notin K$. $\therefore K \cup \{\bar{x}\}$ is lin dep, so $\bar{x} \in \mathrm{span}(K)$, by theorem [$S \cup \{x\}$ is lin ind if & only if $x \in \mathrm{span}(S)$]

$\therefore$, K is lin ind & spans V, $\therefore$ it's a basis with $S \subseteq K$

5. **(extra credit)** For each of the following claims, state whether it is **true or false** (no reason is required).

(a) (2 pts) In any field F, $xz = yz$ implies $x = y$ for any $x, y, z \in F$.

False

(b) (2 pts) The sets $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ and $\{\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}\}$ are both bases of a vector space V.

False

(c) (2 pts) In any vector space, $a\mathbf{x} = b\mathbf{x}$ implies $a = b$ for any $\mathbf{x} \neq \mathbf{0}$ in V and any $a, b \in F$.

True

(d) (2 pt) Any set in a vector space V containing the zero vector $\mathbf{0}_V$ is linearly independent.

False

(e) (2 pt) Suppose that $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is a linearly dependent set in a vector space V and that $\{\mathbf{v}, \mathbf{w}\}$ is a linearly independent set. Then $\mathbf{u}$ is in the span of $\{\mathbf{v}, \mathbf{w}\}$.

True